# Lecture 20

Generative AI / Physical AI

# Vision-Language-Action Models 4: GR00T N1 and Gemini Robotics

I050-4-6338-01 · Spring 2026

Instructor: Prof. Yong-Hoon Choi

# Announcement

# Assignment Submission Reminder

## ❖ Assignment Guide

- Assignment guide repository: https://github.com/KWU-FAIR-LAB/Raccoonbot_Openvla
- New content has been added in Sections 4-5.
- They explain how to move the real Raccoonbot consistently with the simulation setup.

## ❖ Submission

- Due: June 7, 2026 (Sun)
- Submit your GitHub repository URL via the KLAS assignment submission system.
- Include: README.md, modified code, logs, screenshots, and report.pdf.
- Optional: a short demo video.

## ❖ Important Rule

- Do not upload large datasets or model checkpoints. Provide download links if necessary.
- Clearly explain your changes, execution method, and results.

# Final Exam Notice

## ❖ Final Exam

- Date & Time: June 8, 2026 (Mon), 12:00–13:15
- Duration: 75 minutes
- Location: Saebit Hall, Room 104
- Format: Closed-book written exam

## ❖ Exam Coverage

- Lectures 11—20

## ❖ Important Rule

- No textbooks, notes, internet, or other reference materials are allowed.

# Basis for Final Grade

## Grading Components

- Midterm Exam: 20%
- Final Exam: 20%
- Midterm Coding Test: 20%
- Final Project (Assignment): 20%
- Attendance: 20%

## Note

- The final project is the Raccoonbot GitHub assignment announced in class.

# GROOT

# GR00T (Generalist Robot 00 Technology)

## ❖ Paper

- NVIDIA, GR00T N1: An Open Foundation Model for Generalist Humanoid Robots, 2025

## ❖ Main Points

- GR00T N1 is an open VLA foundation model for humanoid robots.
- Dual-system architecture:
  - ✓ System 2: vision-language understanding
  - ✓ System 1: fast continuous action generation
- Trained with heterogeneous data:
  - ✓ real robot trajectories
  - ✓ simulated robot data
  - ✓ human videos
  - ✓ synthetic data
- Updated through GR00T N1 $\rightarrow$ N1.5 $\rightarrow$ N1.7

# Why Humanoid VLA?

## ❖ Why Humanoid Robots Need Foundation Models?

- Humanoids have many degrees of freedom.
- They must operate in human environments.
- Tasks require:
  - ✓ perception
  - ✓ language understanding
  - ✓ whole-body coordination
  - ✓ bimanual manipulation
- Hand-coded skills do not scale to open-ended tasks.

DoF Perspective

- Approx. 44 DoF
  Arms 14 + hands 12 + waist 3 + head 3 + legs 12
- 200+ DoF: toward human-like anatomical motion
  Fine hands, spine, feet, face, and subtle coordination

# What is GR00T N1?

## Inputs

- RGB images
- language instruction
- robot proprioception

## Output

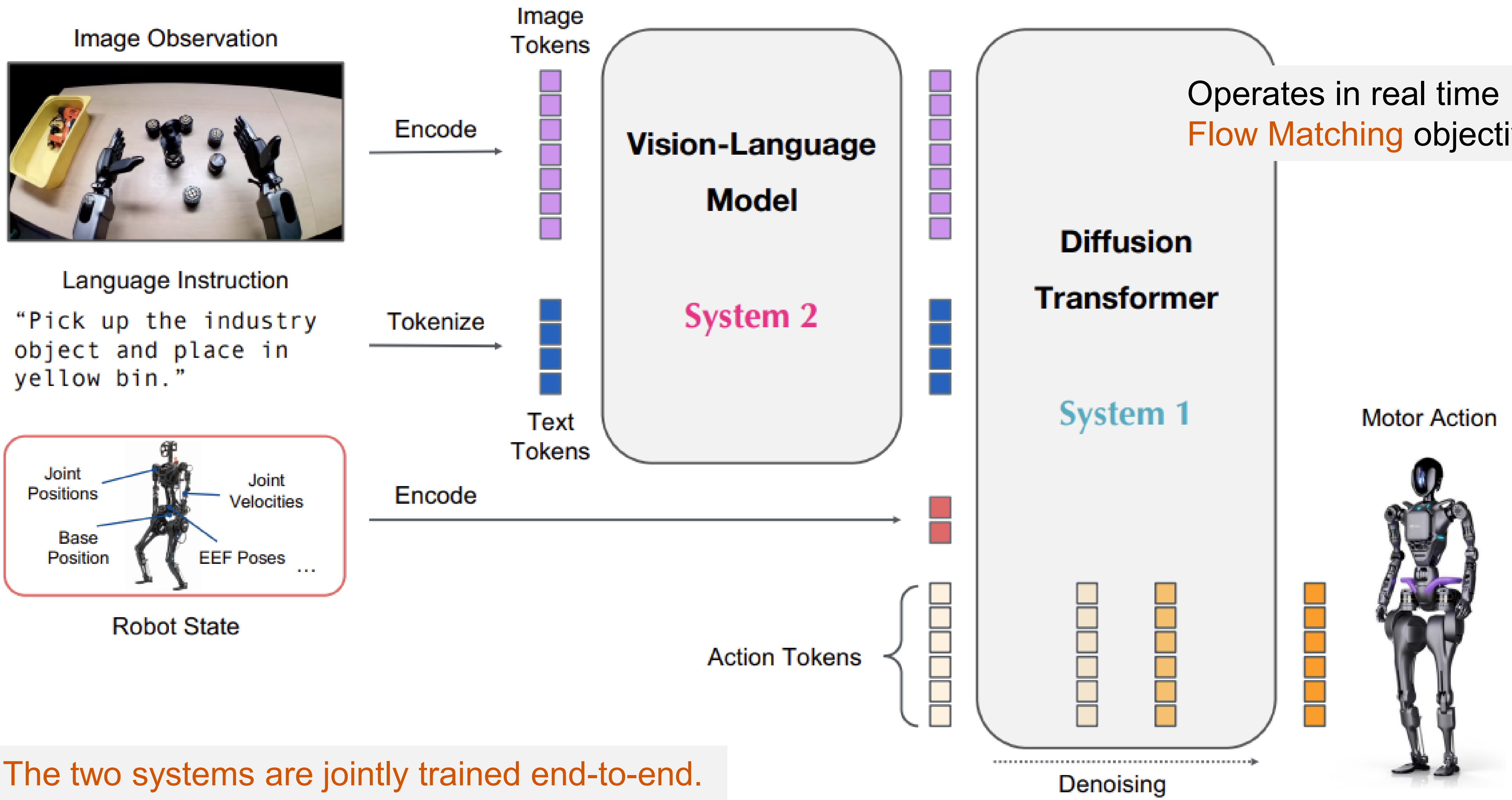
- continuous action chunks for robot control

## Goal

- language-conditioned bimanual manipulation
- generalization across embodiments and tasks

GR00T turns vision and language into humanoid actions.

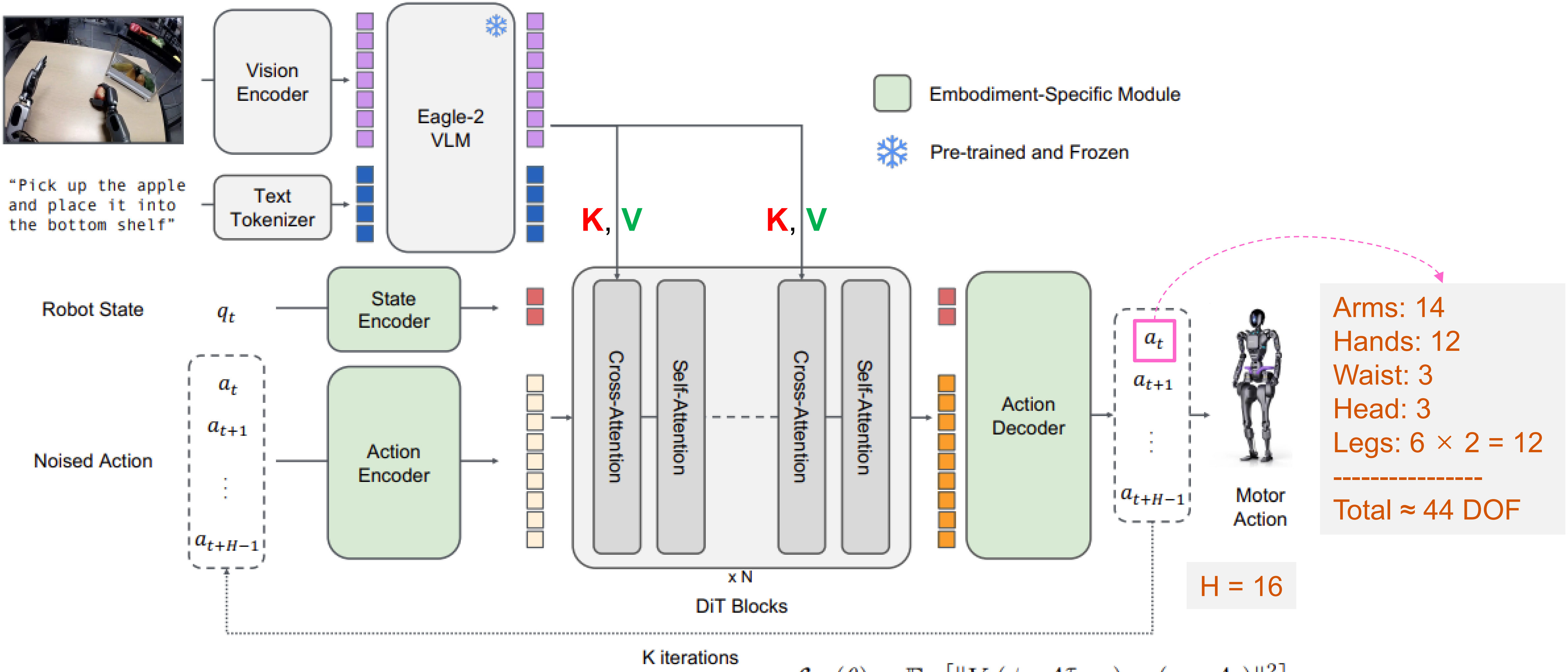
# GR00T N1 Model Overview




Operates in real time
Flow Matching objective

The two systems are jointly trained end-to-end.

# GR00T N1 Model Architecture




$$\mathcal{L}_{fm}(\theta) = \mathbb{E}_{\tau}\left[\|V_\theta(\phi_t, A_t^{\tau}, q_t) - (\epsilon - A_t)\|^2\right]$$

# Training Datasets

## ❖ Data Mixture

- Real humanoid robot teleoperation data
- Cross-embodiment robot data
- human videos
- simulated trajectories
- synthetic / generated data

## ❖ Key Points

- Large-scale humanoid data is expensive and scarce.
- GR00T overcomes this by combining heterogeneous data sources.
- Humanoid foundation models need more than humanoid teleoperation data.

# Result


# Isaac-GR00T GitHub Repository

## ❖ NVIDIA/Isaac-GR00T

- Official GitHub repository for GR00T N1.7
- Open VLA model for generalized humanoid robot skills
- Provides:
  - ✓ pre-trained model weights
  - ✓ reference code
  - ✓ inference workflow
  - ✓ fine-tuning with custom robot data
  - ✓ evaluation and deployment examples
- Uses GR00T LeRobot-format data
- Supports post-training for new embodiments, tasks, and environments

## ❖ Links

- Pretrained model: https://huggingface.co/nvidia/GR00T-N1.7-3B

# Gemini Robotics

# Gemini Robotics

## ❖ Paper

- Google DeepMind, Gemini Robotics: Bringing AI into the Physical World, 2025
- https://deepmind.google/models/gemini-robotics/

## ❖ Main Points

- Gemini Robotics is a generalist VLA model for robot manipulation.
- Built on the Gemini 2.0 foundation model.
- Two main components:
  - ✓ Gemini Robotics-ER: embodied reasoning
  - ✓ Gemini Robotics: action generation
- Key capabilities:
  - ✓ dexterous manipulation
  - ✓ instruction following
  - ✓ visual, instruction, and action generalization

# Motivation: From Digital AI to Physical AI

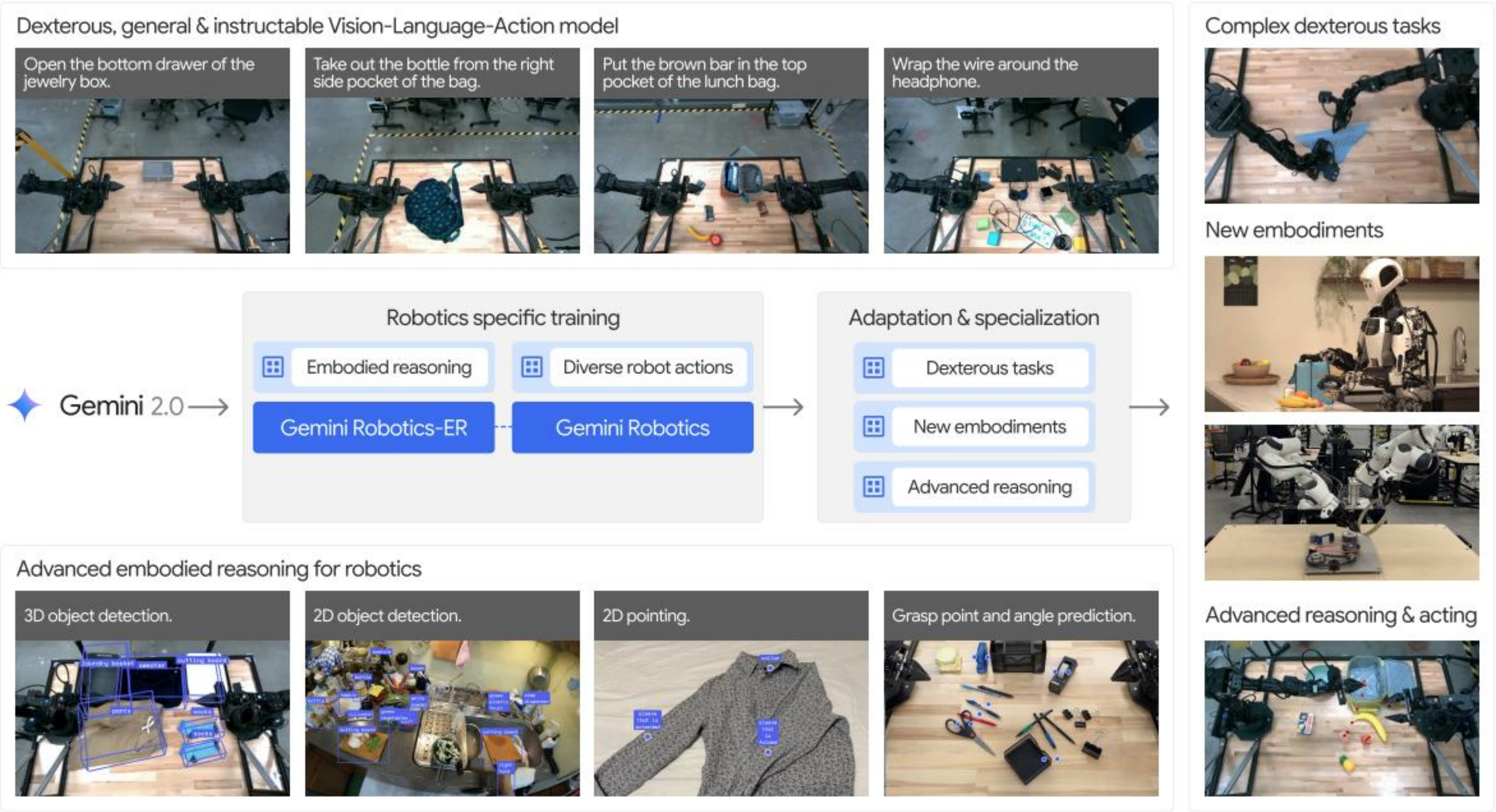
## ❖ Motivation

- Large multimodal models have achieved strong capabilities in digital domains.
- However, robots must operate in the physical world.
- Physical AI requires more than image and language understanding.
- Robots need to reason about:
  - ✓ 3D spatial structure
  - ✓ object relationships
  - ✓ affordances
  - ✓ contact and motion
  - ✓ safe physical actions

## ❖ Key Question

- How can Gemini's multimodal intelligence be grounded in real-world robot actions?

# Gemini Robotics Family

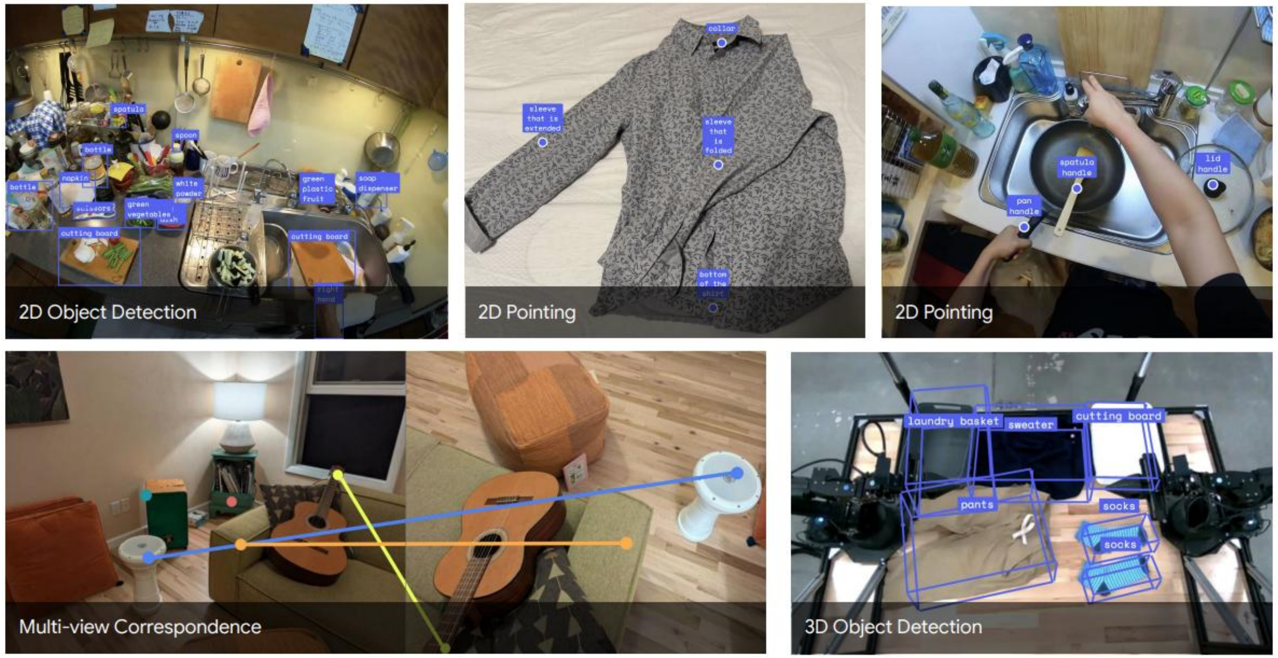


## ❖ What is Embodied Reasoning?

- Embodied reasoning is physical-world reasoning for action.
- It connects visual understanding with robot-relevant decisions.
- The model must understand not only objects, but also how to interact with them.

## ❖ Examples

- Detect objects and parts.
- Point to relevant locations.
- Predict grasp points.
- Predict motion trajectories.
- Understand 3D spatial relationships.
- Reason about task-relevant states.



## ❖ Reasoning for Physical Interaction

- Robotics requires reasoning beyond object recognition.
- A robot must understand motion consequences, tool use, and spatial relations.

## ❖ Example Questions from ERQA

| Trajectory Reasoning | Action Reasoning | Spatial Reasoning |
|---|---|---|
|  |  |  |
| Approximately which colored trajectory should the zipper follow to begin zipping up the suitcase?<br><br>**A. Blue**<br>B. Purple<br>C. Green<br>D. Red | How should the person move the wrench so that it is ready to rotate the hex screw closest to it?<br><br>**A. Forward and right**<br>B. Up and left<br>C. Forward and left<br>D. None of the above | There are 4 sinks in the picture. Which arrow points to the one that is closest to the viewer?<br><br>A. Cyan<br>B. Blue<br>C. Red<br>**D. None of the arrows** |

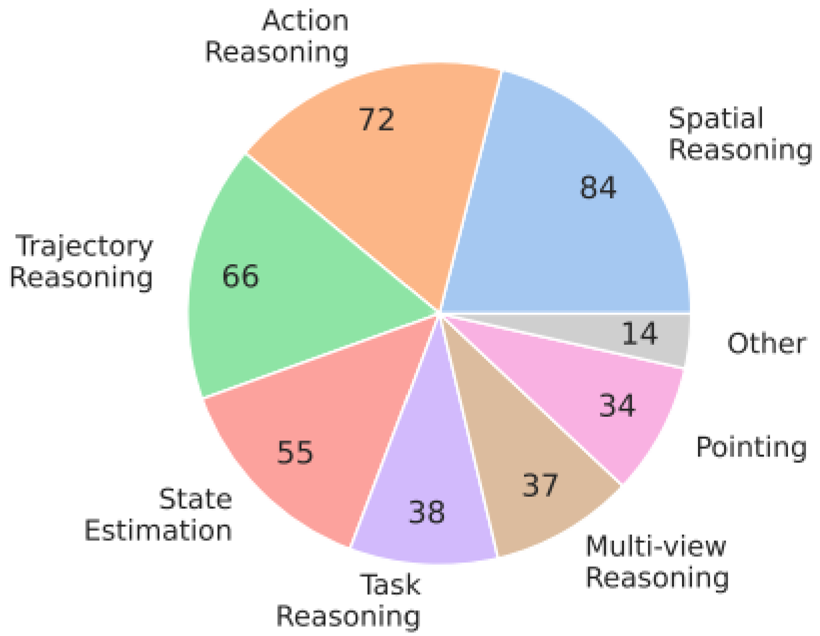
# ERQA Benchmark

## ❖ Embodied Reasoning Question Answering (ERQA)

- ERQA is introduced as an evaluation benchmark for embodied reasoning.
- It contains 400 multiple-choice VQA-style questions.

## ❖ Question Categories

- Spatial reasoning
- Action reasoning
- Trajectory reasoning
- State estimation
- Task reasoning
- Multi-view reasoning
- Pointing




## ❖ Key Message

- ERQA evaluates embodied reasoning, not robot action generation.

# ERQA Benchmark Results

## ❖ Benchmark Results

- Gemini 2.0 models are evaluated on ERQA, RealWorldQA, and BLINK.
  - ✓ ERQA is the most challenging benchmark among the three.
- Gemini 2.0 Pro Experimental achieves the best ERQA score.

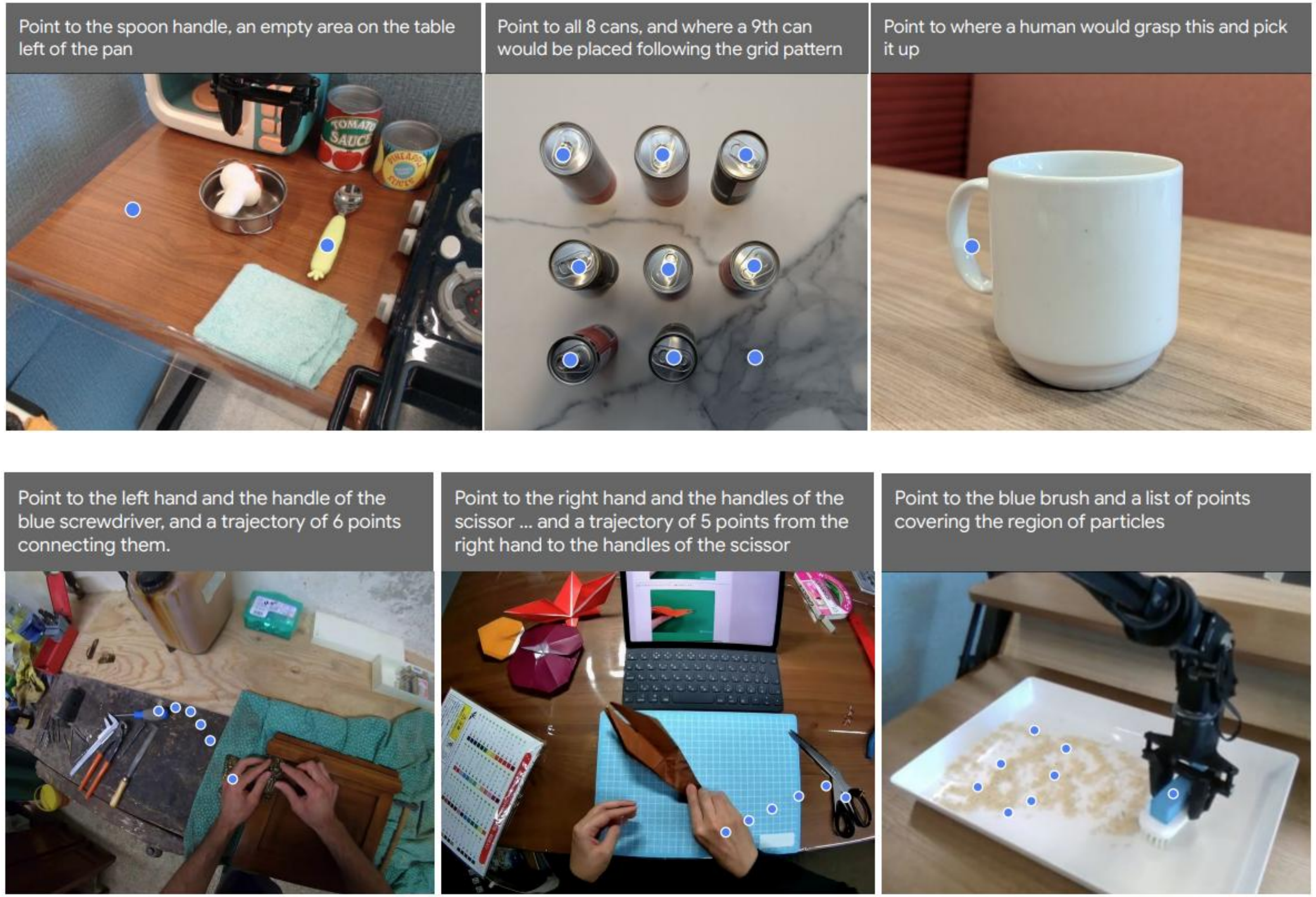
| | Gemini | | | | GPT | | Claude |
|---|---|---|---|---|---|---|---|
| Benchmark | 1.5 Flash | 1.5 Pro | 2.0 Flash | 2.0 Pro Experimental | 4o-mini | 4o | 3.5 Sonnet |
| ERQA | 42.3 | 41.8 | 46.3 | **48.3** | 37.3 | 47.0 | 35.5 |
| RealworldQA (test) | 69.0 | 64.5 | 71.6 | **74.5** | 65.0 | 71.9 | 61.4 |
| BLINK (val) | 59.2 | 64.4 | 65.0 | **65.2** | 56.9 | 62.3 | 60.2 |

- With CoT prompting, Gemini 2.0 Pro improves from 48.3 → 54.8.

## ❖ Key Message

- Step-by-step reasoning improves embodied reasoning performance.

# Results: Pointing and Trajectories




숟가락 손잡이와 팬 왼쪽 테이블의 빈 공간을 올바르게 가리킴

8개 캔의 위치와 9번째 캔을 놓을 위치를 예측

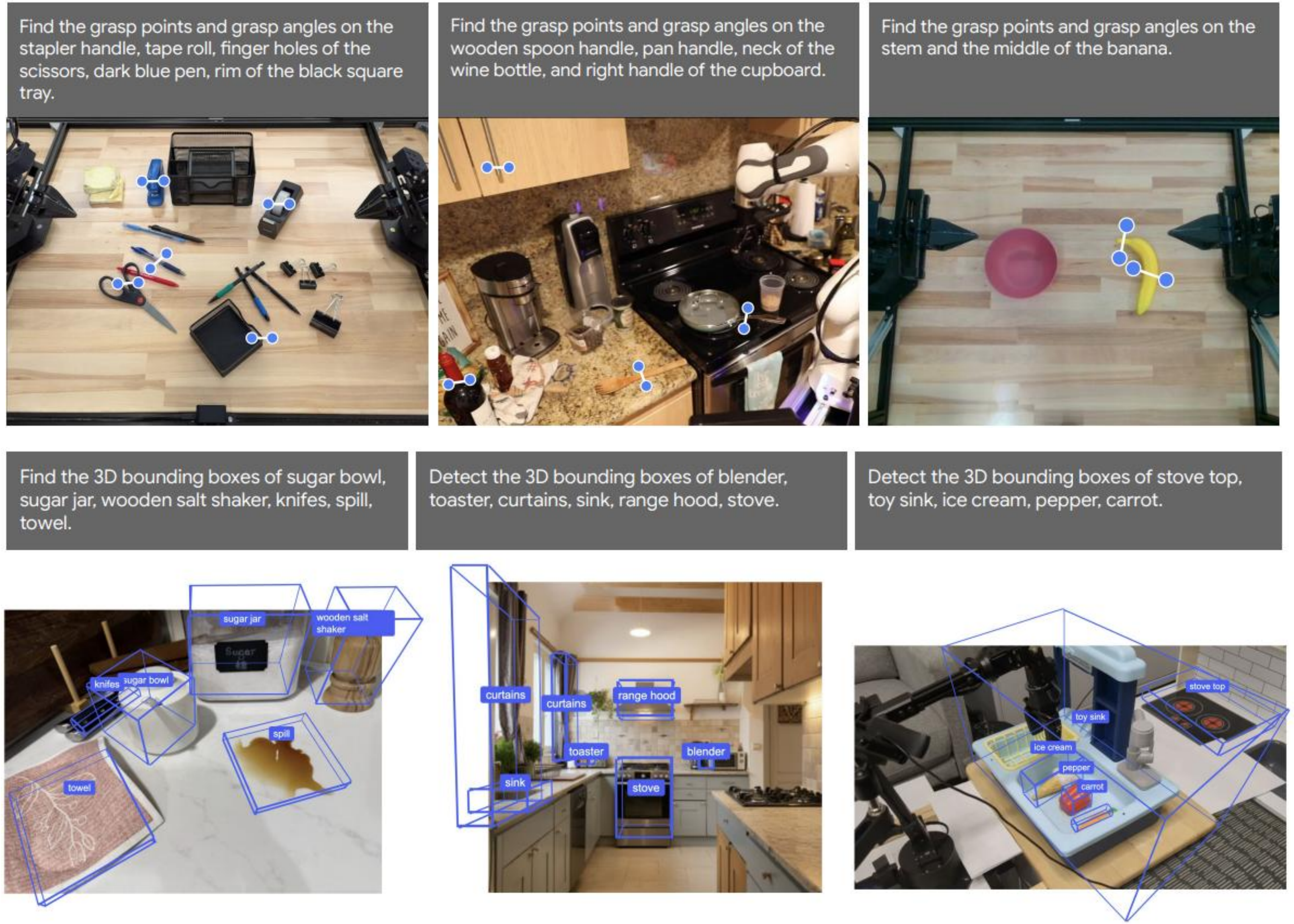
사람이 컵을 집기 좋은 손잡이 위치를 가리킴

왼손에서 드라이버 손잡이까지의 이동 경로를 예측

오른손에서 가위 손잡이까지의 이동 경로를 예측

브러시와 닦아야 할 입자 영역을 함께 가리킴

# Results: Grasp Prediction and 3D Detection




여러 물체의 집기 위치와 그리퍼 각도 예측

주방 물체의 손잡이·목 부분 등 잡을 위치 예측

바나나의 줄기와 몸통 집기 위치를 구분하여 예측

작은 물체와 액체 영역을 3D 박스로 검출

주방 장면에서 가전·가구의 3D 위치와 크기 추정

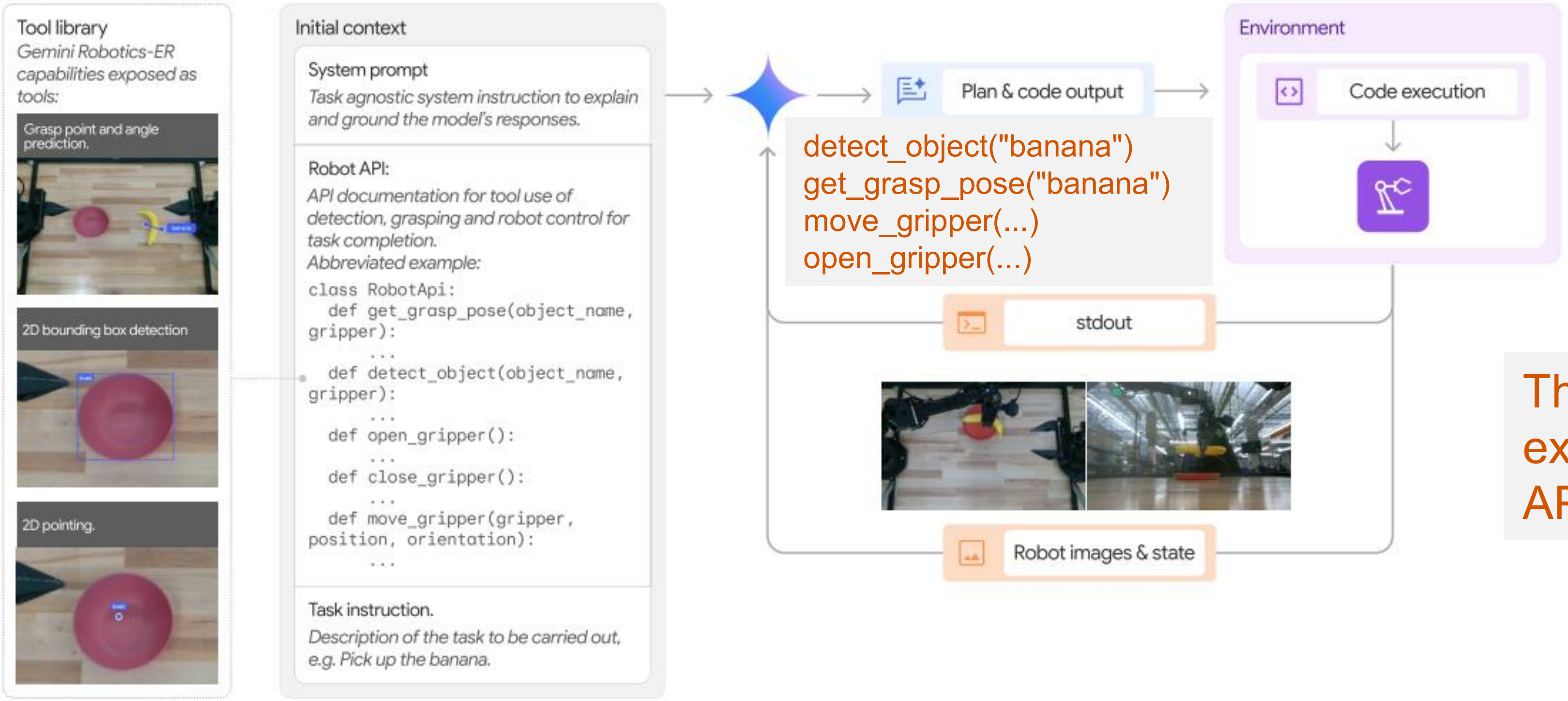
장난감 환경에서 물체별 3D 경계 박스 예측

# Gemini Robotics-ER: Indirect Robot Control

## ❖ Zero-Shot Control via Code Generation

- Input: task instruction, robot images, robot state, and robot API
- Output: executable robot control code
- Feedback enables error detection and re-planning.

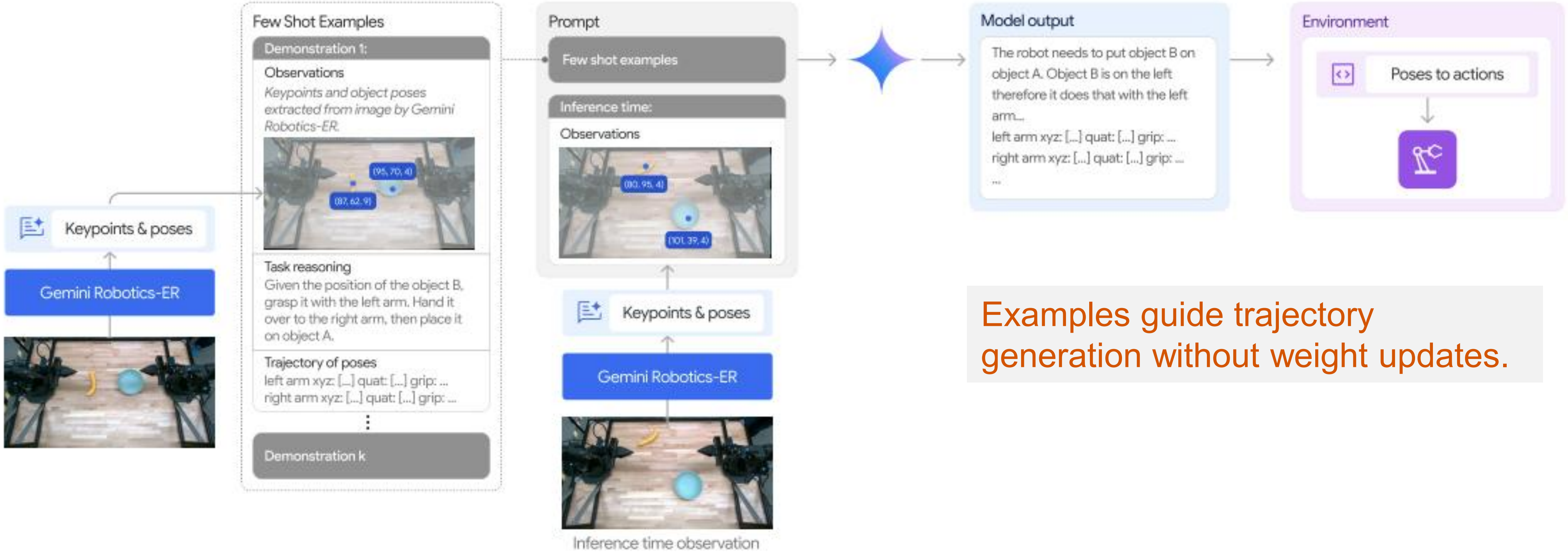
Gemini Robotics-ER는 로봇 액션을 직접 출력하는 VLA는 아니다. 대신 로봇 API를 이해하고, 현재 이미지와 상태를 보고, 다음에 실행할 코드를 생성한다. 실행 후 feedback을 받아 실패하면 다시 계획한다.




The generated code is executed through robot APIs to move the robot.

## ❖ Few-Shot Robot Control

- Demonstrations are provided in the prompt.
- The model predicts end-effector trajectories for new task instances.
- No weight update – the model adapts its output pattern from the context.

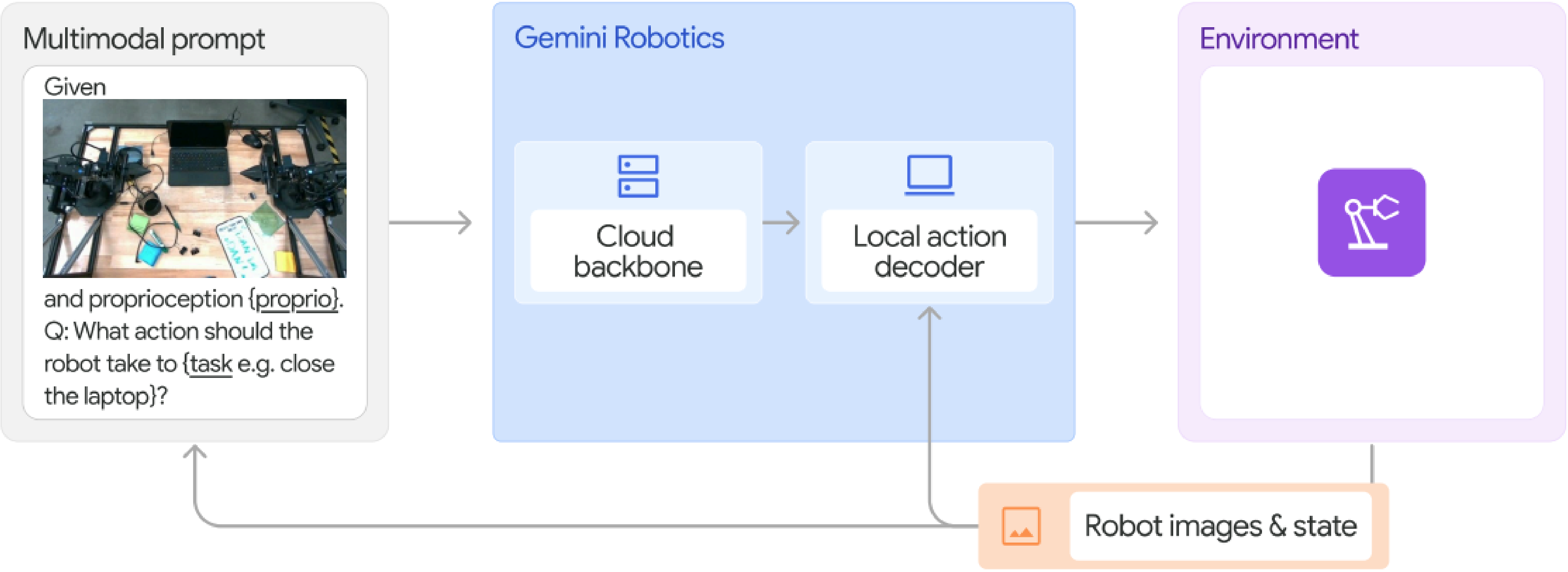



Examples guide trajectory generation without weight updates.

# Gemini Robotics: Direct VLA Control

## ❖ From Indirect Control to Direct VLA Control

- Gemini Robotics-ER controls robots indirectly through code or trajectory generation.
- Gemini Robotics is fine-tuned with robot action data.
- It directly predicts action chunks from images, instructions, and proprioception.
- The generated action chunks are executed by the robot.



# Control Approaches in Gemini Robotics

## ❖ Three Separate Control Approaches

| Approach | Model | Model Output | Weight Update |
|---|---|---|---|
| Zero-shot control | Gemini Robotics-ER | Robot control code | No |
| Few-shot in-context control | Gemini Robotics-ER | End-effector trajectories | No |
| Direct VLA control | Gemini Robotics | Robot action chunks | Yes, during training |

## ❖ Key Distinction

- Gemini Robotics-ER uses embodied reasoning for indirect robot control.
- Gemini Robotics directly predicts action chunks as a VLA policy.
- Code generation is not converted into action chunks.

## ❖ Key Message

- These are separate control approaches, not stages of one pipeline.

# Training Data for Gemini Robotics

## ❖ Robot Action Data

- Gemini Robotics is trained with large-scale robot action data.
- The data was collected from a fleet of ALOHA 2 robots.
- It includes thousands of hours of real-world teleoperated demonstrations.
- The demonstrations cover diverse manipulation tasks, objects, and environments.

## ❖ Additional Multimodal Data

- web documents
- code
- images, audio, and video
- embodied reasoning data
- visual question answering data

https://aloha-2.github.io/

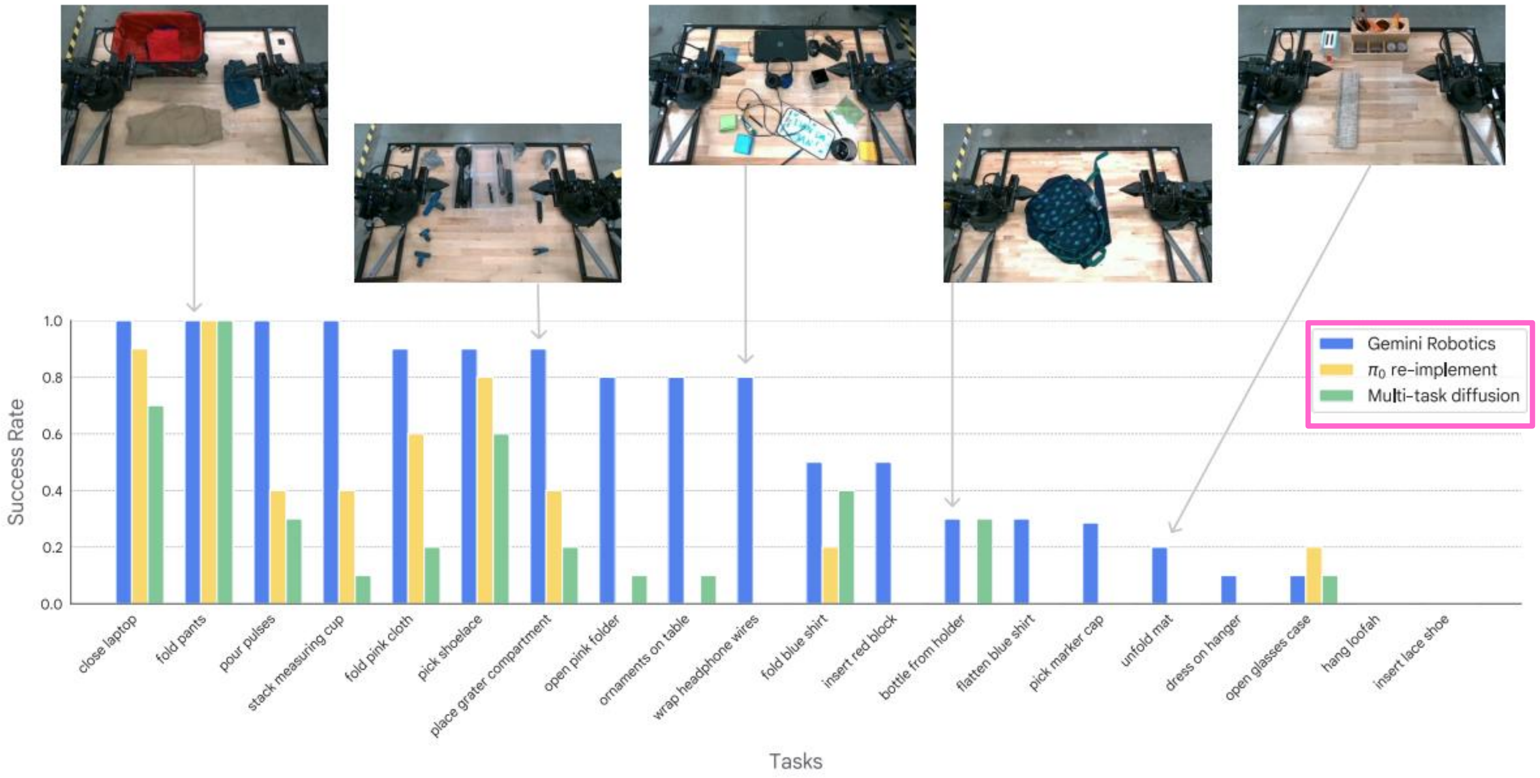
## ❖ Training Objectives

- Exact action training objective is not disclosed in the report.

# Results: Diverse Dexterous Manipulation

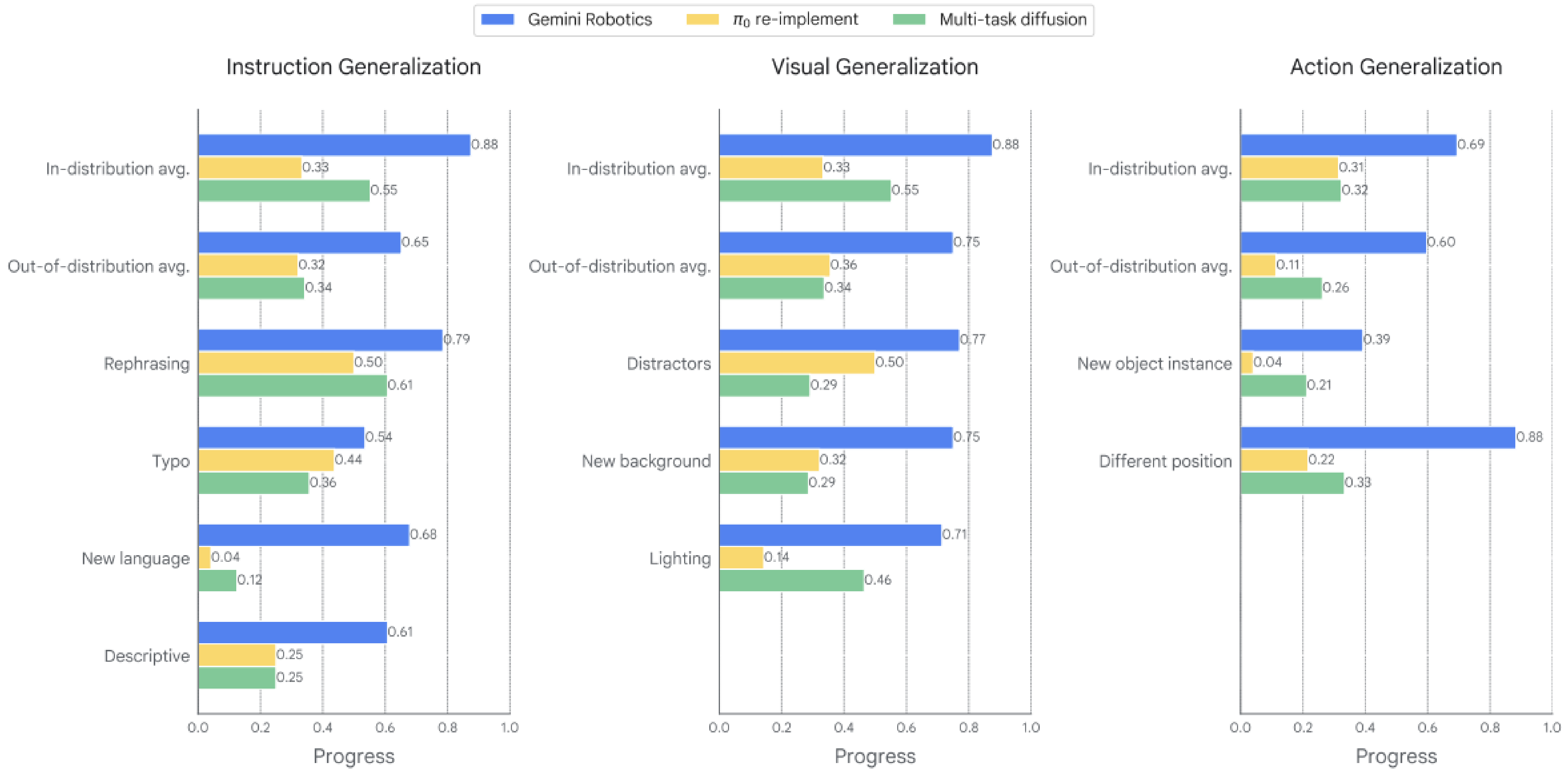
## ❖ Out-of-the-Box Manipulation

- Evaluated on 20 short-horizon dexterous tasks
- No task-specific fine-tuning
- Covers pick-and-place, deformable objects, and bimanual manipulation




Gemini Robotics outperforms baselines across diverse manipulation tasks.

# Generalization Capabilities

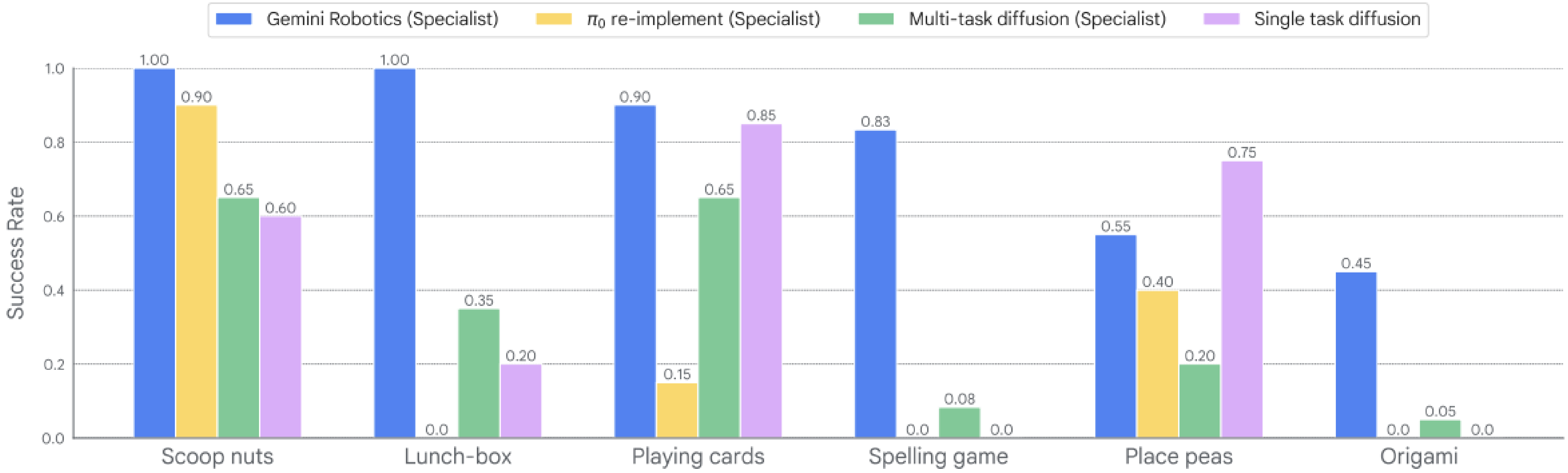



Can the robot act correctly under unseen instructions, visual changes, and object variations?

# Specialization for Long-Horizon Dexterity

## ❖ Specialization

- Fine-tuned for challenging long-horizon dexterous tasks
- Requires precise bimanual coordination and fine manipulation
- Examples: origami, lunch-box packing, card game, spelling game

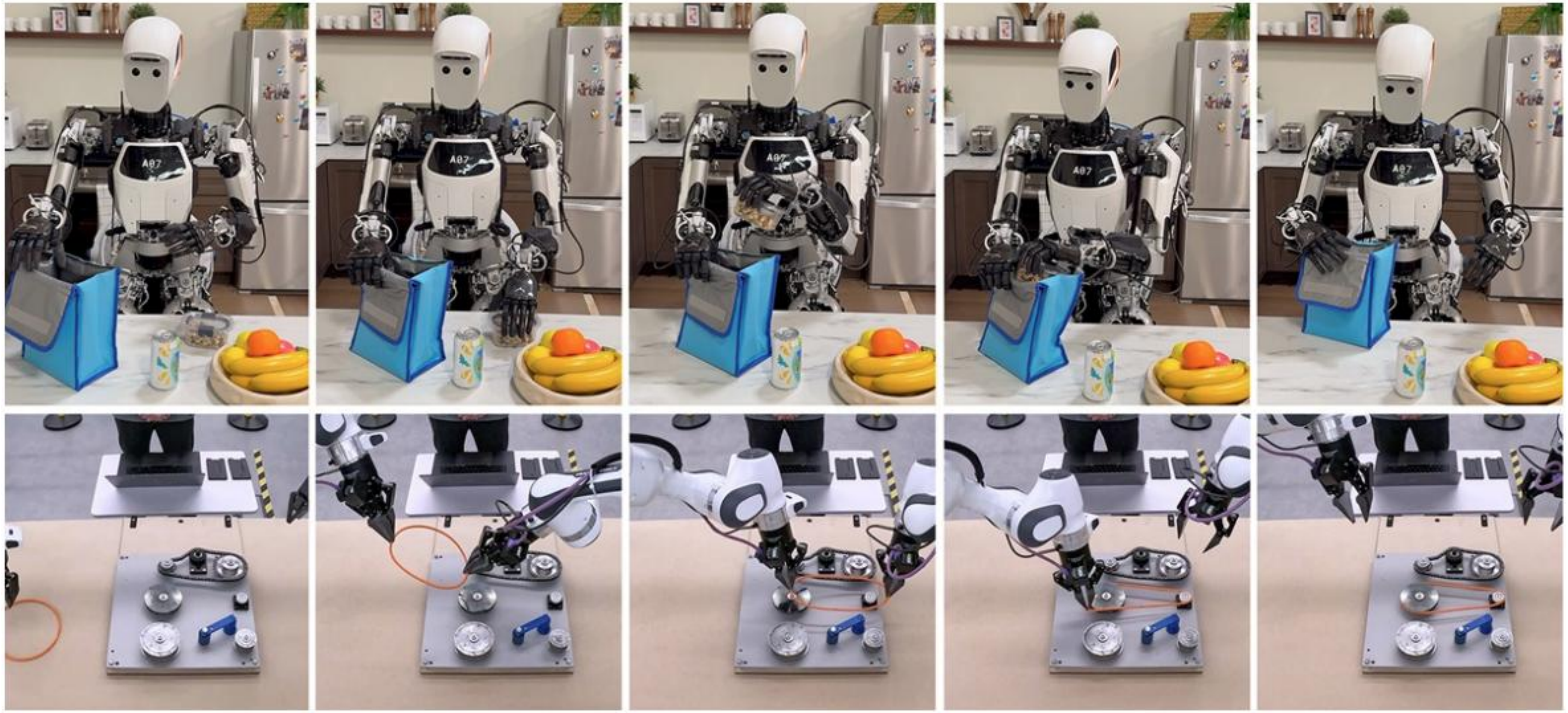



A generalist VLA can become a strong specialist through fine-tuning.

# Adaptation to New Robot Embodiments

## ❖ New Robot Embodiments

- Fine-tuned from an ALOHA-2-trained model to different robot platforms.
- Demonstrated on Apollo humanoid and bi-arm Franka.



# Safety Considerations

## ❖ Safety

- Robots act in the physical world.
- VLA models must be combined with safety constraints and low-level controllers.
- Unsafe instructions and risky physical actions must be detected and avoided.

## ❖ Limitations

- Controlled experimental settings.
- Additional data required for specialization and new embodiments.
- Safe deployment remains an open challenge.

## ❖ Key Messages

- A powerful VLA policy is not sufficient by itself for safe robot deployment.

**자동차가 도로·법규·신호체계·안전장치(에어백, ABS)·보험·교통경찰과 함께 배포되듯, 로봇도 VLA 모델만으로는 충분하지 않다.**

# Final Takeaways

## ❖ Key Takeaways

- Gemini Robotics extends Gemini 2.0 from digital intelligence to physical AI.
- Gemini Robotics-ER provides embodied reasoning for physical-world understanding.
- Gemini Robotics directly predicts robot action chunks as a VLA policy.
- The model shows strong manipulation, generalization, and embodiment adaptation.

## ❖ Overall Message

- Generalist VLA models are a major step toward general-purpose robots.
- However, real-world deployment still requires data, embodiment adaptation, and safety ecosystems.

Thank you for your curiosity and effort throughout this journey from digital intelligence toward the physical world.